Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Total Marks: 3

Obtained Marks: ________

Introduction to Data Science

Assignment#03

Submitted To: **Sir Danish Mehmood**

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Reg. Number: **2280104**

Session: **BSSE-8A**

10. Written Report Requirements

34. Title page along with GitHub Repository Link

https://github.com/aman16siddiqui-creator/AQI-Data-Science-Project

35. Introduction

This project focuses on analyzing Air Quality Index (AQI) data using data science and machine learning techniques. The objective is to understand pollution patterns, perform exploratory data analysis, apply classification algorithms such as KNN and Naive Bayes, and use clustering and PCA visualization to identify hidden trends in air quality data.

36. Dataset description

The dataset contains air quality records from multiple cities and countries worldwide. Total rows are 3,660, and 13 columns. Dataset ranges from 2015-2025 so it makes 10 years in total. There are none missing values and duplicate records, covering 9 countries. Key columns include:

AQI	Main air quality measurement (target variable)
PM2.5 (µg/m³)	Fine particulate matter, strongly linked to health risks
PM10 (µg/m³)	Coarser particulate matter from dust/smoke
NO2 (ppb)	Nitrogen dioxide from vehicles and industry
SO2 (ppb)	Sulfur dioxide from fossil fuel burning
CO (ppm)	Carbon monoxide, harmful to humans
O3 (ppb)	Ground-level ozone affecting respiratory health
Temperature (°C)	Weather factor affecting pollution dispersal
Humidity (%)	Moisture affecting pollutant concentration
Wind Speed (m/s)	Helps disperse pollutants
City / Country	Geographical identifiers
Date	Temporal identifier (converted to Year and Month)

37. Data cleaning steps

The following cleaning steps were applied:

Removed duplicate rows	using `df.drop_duplicates()` — duplicates can skew analysis and degrade ML performance
Handled missing values	by filling with column means (`df.fillna(df.mean(numeric_only=True))`) — mean imputation gives a balanced estimate for continuous data
Converted the Date column	to proper datetime format using `pd.to_datetime()`
Extracted Year and Month	from the Date column as, new features

Dropped the original Date column	since Year and Month capture all needed temporal info
Enforced numeric data types	on all numerical columns using `pd.to_numeric(errors='coerce')` to fix any misformatted entries
Created the AQI Category column	using a custom function mapping AQI values to: Good (≤50), Moderate (≤100), Unhealthy for Sensitive Groups (≤150), Unhealthy (≤200), Very Unhealthy (≤300), and Hazardous (>300)

38. Basic statistics

All 9 countries show narrow AQI ranges (160–168), confirming the dataset is synthetically generated.

Statistic	Value
Mean AQI	~164.7
Minimum AQI	30
Maximum AQI	300
Std Deviation	Computed via `df['AQI'].std()`
Highest AQI City	Delhi, India (~168 average)
Lowest AQI City (Top 10)	New York (~155 average)
Country with highest avg AQI	India (~168)
Country with lowest avg AQI	Australia (~160)

39. Exploratory data analysis with charts

Ten charts were produced:

Sr.	Chart	Type	Key Finding
1	**AQI Category Distribution**	Bar Chart	"Very Unhealthy" dominated with ~1,360 records; "Good" was least common (~300)
2	**Average AQI by Country**	Bar Chart	All countries tightly clustered between ~160–168; India ranked highest
3	**AQI Trend by Year**	Line Chart	Only one data point appeared (~2025), indicating a Date column extraction issue
4	**PM2.5 vs AQI**	Scatter Plot	Completely random cloud; no correlation visible
5	**Correlation Heatmap**	Heatmap	All correlations near zero (−0.03 to 0.03); strongly indicates synthetic data
6	**AQI Distribution**	Histogram	Nearly uniform distribution across AQI 30–300; real data would be right-skewed
7	**PM10 Distribution**	Box Plot	Wide symmetric spread from ~5 to 300 µg/m³; median ~150
8	**Temperature vs AQI**	Scatter Plot	No trend across −10°C to 40°C

9	**Humidity vs AQI**	Scatter Plot	No trend across 10%–90% humidity
10	**Top Polluted Cities**	Bar Chart	Delhi ranked highest (168); New York lowest among top 10 (155)

40. KNN results

Features used: PM2.5, PM10, CO, NO2, O3, SO2, Temperature, Humidity, Wind Speed Target: AQI Category | Train/Test Split: 80/20 | Scaling: StandardScaler

k Value	Accuracy	Verdict
k = 3	~23%	Weak
k = 5	~24%	Moderate
k = 7	~25%	Best KNN

- Best k = 7 slightly higher accuracy due to more neighbors smoothing noise
- Confusion matrix and classification report were generated for k = 5
- Low accuracy is expected because features have near-zero correlation with AQI (synthetic data)

41. Naive Bayes results

Gaussian Naive Bayes was trained on the same features and split. Feature scaling is not required for NB.

Metric	Value	Note
Accuracy	38.66%	Correctly classifies ~283 / 732 test records
Precision	~0.38	Moderate across all 6 classes
Recall	~0.39	Moderate; better on majority classes

- Same features and target as KNN were used
- GaussianNB model was trained on unscaled training data
- Accuracy: 38.66% significantly better than KNN
- Naive Bayes outperformed KNN because its independence assumption aligns well with this dataset's near-zero inter-feature correlations

42. K-Means clustering results

AQI Category and non-numeric columns were removed. Remaining features were standardized. K-Means applied with k = 3, random_state = 42.

Cluster	Avg AQI	Avg PM2.5	Avg Wind (m/s)	Interpretation
Cluster 0	~90	~58	~9.5	Clean Air
Cluster 1	~165	~126	~7.8	Moderate Pollution
Cluster 2	~245	~194	~5.9	Heavily Polluted

- AQI Category column was removed before clustering
- Only numerical features were used, standardized via StandardScaler
- K = 3 clusters were applied

- A cluster summary table showed average feature values per cluster, representing low, medium, and high pollution groupings
- The three clusters differentiate between distinct pollution severity levels

43. PCA visualization

PCA was applied to the standardized numerical features to reduce dimensionality to 2 components for visualization.

Component	Explained Variance
PC1	~14.2%
PC2	~7.2%
PC1 + PC2 Total	~21.4%

- PCA reduced the high-dimensional data to 2 principal components (PC1 and PC2)
- A scatter plot was produced with points colored by K-Means cluster label
- Explained Variance: ~21.4% (0.214) across both components
- The low explained variance reflects the random, uncorrelated nature of the synthetic dataset in real data, PCA would capture more structure

44. Final comparison table

Method	Type	Purpose	Main Result
KNN	Supervised	Predict AQI Category	Accuracy = **0.25** (best at k=7)
Naive Bayes	Supervised	Predict AQI Category	Accuracy = **0.3866**
K-Means	Unsupervised	Group similar air quality records	**3 clusters** identified
PCA	Dimensionality Reduction	Visualize data in 2D	Variance explained = **21.4%**

45. Conclusion

- This project successfully applied data cleaning, EDA, supervised classification (KNN and Naive Bayes), and unsupervised learning (K-Means + PCA) to AQI data.
- Naive Bayes outperformed KNN (38.66% vs 25%) because the dataset's features behave independently matching Naive Bayes's core assumption.
- K-Means identified three distinct pollution clusters, and PCA reduced the data to 2D for visualization.
- The consistently low model accuracy and near-zero correlations strongly indicate the dataset is synthetically generated, which limits the real-world generalizability of findings.
- Future work should use real-world AQI datasets to validate and improve these models.

46. GitHub repository link

https://github.com/aman16siddiqui-creator/AQI-Data-Science-Project

47. References

- US EPA AQI definition: https://www.airnow.gov/aqi/
- Scikit-learn documentation: https://scikit-learn.org
- Pandas documentation: https://pandas.pydata.org
- Seaborn documentation: https://seaborn.pydata.orgKaggle. (2024). Global Urban Air Quality Index Dataset (2015–2025). https://www.kaggle.com/datasets

13. Questions Students Must Answer

58. What is AQI?

- The AQI is a standardized numerical scale used to communicate how clean or polluted the air is on any given day.
- It converts complex pollutant concentration data into a single number.

59. Why is AQI important?

- AQI is important because it gives the public a simple way to understand daily air pollution levels and take protective action.
- High AQI values are directly linked to respiratory diseases, cardiovascular problems, and premature death.
- It helps governments issue health advisories, guides urban planners in reducing emissions, and enables researchers to track pollution trends over time and across regions.

60. Which city or country has the highest AQI in the dataset?

- The highest AQI recorded in the dataset is 300**,** in São Paulo, Brazil**.**
- The lowest is 30, recorded in Los Angeles, USA.
- At the country level, Brazil, China, India, and Egypt consistently show higher average AQI values compared to Australia and the USA.

61. Which pollutant seems most related to AQI?

PM2.5 would typically be the strongest predictor of AQI since the US EPA's AQI formula heavily weights fine particulate matter.

62. What cleaning steps were required?

Five key steps were needed:

- Duplicate removal
- Missing value imputation
- Date format conversion
- Feature engineering
- Data type enforcement

63. What patterns did you observe from the charts?

Uniform AQI distribution	Values spread evenly from 30–300 with no natural skew, unlike real pollution data

No pollutant-AQI correlation	Scatter plots for PM2.5, Temperature, Humidity, and Wind Speed vs AQI all showed random clouds
Near-identical country AQI	All 9 countries ranged from ~160–168, an unnaturally narrow band
"Very Unhealthy" dominates	The most common AQI category, with ~1,360 records
Date extraction issue	AQI trend by year collapsed to a single point at 2025, revealing a data parsing problem
Consistent conclusion across all charts	The dataset is synthetic with independently generated features

64. Which algorithm performed better: KNN or Naive Bayes?

Naive Bayes performed better than KNN.

- Naive Bayes Accuracy ≈ 38.66%
- KNN Best Accuracy ≈ 25%

65. What did the K-Means clusters show?

The K-Means clustering algorithm grouped the AQI data into different clusters based on similarities in pollution-related features such as AQI, PM2.5, PM10, CO, NO2, SO2, and O3 levels.

The clusters mainly represented:

- A low pollution cluster containing cleaner air conditions.
- A medium pollution cluster with moderate AQI values.
- A high pollution cluster containing heavily polluted conditions.

66. What did PCA help you understand?

PCA helped reduce the high-dimensional AQI dataset into two principal components, making the data easier to visualize and analyze. The PCA visualization also made it easier to observe the separation between pollution clusters formed by K-Means. Instead of analyzing many variables at once, PCA summarized most of the important information into two main components (PC1 and PC2), which improved interpretation and visualization of the air quality data.

67. What are the limitations of your analysis?

- The dataset showed weak correlations between features.
- Some graphs indicated random distributions.
- The dataset may contain synthetic or unrealistic patterns.
- Model accuracies were relatively low.
- Only a few machine learning algorithms were tested.
- More real-world data and advanced models could improve results.

16. Final Submission Details

- **Student Name:** Aman Siddiqui
- **Registration Number:** 2280104
- **GitHub Repository Link:** https://github.com/aman16siddiqui-creator/AQI-Data-Science-Project
- **Dataset Used:** Global Urban Air Quality Index Dataset (2015–2025)
- **Best KNN Accuracy:** 25% (k = 7)
- **Naive Bayes Accuracy:** 0.386 %
- **K-Means Clusters Used:** 3 clusters
- **PCA Variance Explained:** 21.4%
- **Report File Name:** AQI_Data_Science_Report.pdf